

GEOCHANG, Korea, Republic of

WMO: 471570

Lat: 35.6674N

Lon: 127.9100E

Elev: 741

StdP: 14.31

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	15.3	-6.3	4.0	25.7	-2.1	5.1	28.5	14.4	34.1	12.9	35.0	2.8	250	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.8	91.1	76.0	88.6	74.8	86.0	73.3	78.4	86.6	77.4	84.9	76.4	83.0	4.9	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.4	141.6	81.4	75.4	136.9	80.4	74.5	132.5	79.8	42.6	86.9	41.5	85.3	40.5	83.4	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.4	11.7	10.1	7.1	94.1	3.4	1.6	4.6	95.2	2.6	96.1	0.7	97.0	-1.8	98.1		
(5)			6.0	80.2	3.4	1.4	3.6	81.2	1.6	82.0	-0.3	82.8	-2.7	83.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	54.5	29.5	34.6	43.2	53.1	63.5	71.0	76.9	77.4	68.4	57.1	45.3	33.4	(6)
(7)		DBStd	17.42	6.21	6.61	6.54	6.03	4.66	3.65	4.21	4.58	4.86	5.67	6.90	6.30	(7)
(8)		HDD50	2038	634	434	230	38	0	0	0	0	0	10	176	516	(8)
(9)		HDD65	4922	1099	852	677	358	84	3	0	0	20	255	593	981	(9)
(10)		CDD50	3696	0	3	18	131	418	630	835	849	551	228	33	0	(10)
(11)		CDD65	1105	0	0	0	1	37	183	370	384	121	9	0	0	(11)
(12)		CDH74	8718	0	0	1	68	615	1295	2926	3120	668	25	0	0	(12)
(13)		CDH80	2710	0	0	0	4	120	314	1014	1151	106	1	0	0	(13)
(14)	Wind	WSAvg	3.4	3.5	3.7	4.2	4.3	3.9	3.3	3.0	2.8	2.6	2.8	3.1	3.6	(14)
(15)	Precipitation	PrecAvg	49.8	1.1	1.6	2.2	3.8	3.4	6.7	11.5	9.7	5.4	2.0	1.6	0.8	(15)
(16)		PrecMax	77.1	5.4	4.2	4.0	7.7	6.5	15.0	22.0	18.3	14.8	5.7	3.6	1.9	(16)
(17)		PrecMin	27.1	0.0	0.0	0.2	1.2	0.2	0.1	3.1	1.8	0.2	0.0	0.2	0.0	(17)
(18)		PrecStd	13.0	1.2	1.3	0.9	2.1	1.7	3.7	5.6	4.8	3.7	1.6	1.1	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.2	60.2	71.0	79.7	87.8	89.5	92.9	94.1	86.4	77.2	68.9	55.6	(19)
(20)			MCWB	40.4	52.2	53.1	56.7	62.4	67.9	77.0	77.2	73.4	64.2	58.9	48.8	(20)
(21)		2%	DB	46.7	55.4	65.6	75.6	83.2	85.9	90.8	91.8	83.0	73.2	63.8	50.6	(21)
(22)			MCWB	37.7	46.5	49.1	56.3	61.2	67.5	76.8	77.1	70.8	61.2	54.8	42.2	(22)
(23)		5%	DB	43.2	51.1	61.1	71.6	80.1	83.5	88.3	89.5	80.3	70.7	60.6	47.6	(23)
(24)			MCWB	35.6	40.7	46.9	53.3	60.1	66.9	75.6	76.3	68.7	59.9	52.9	41.0	(24)
(25)		10%	DB	40.4	47.0	56.8	67.4	77.5	81.3	85.7	86.5	78.0	68.1	57.3	44.1	(25)
(26)			MCWB	33.4	37.9	46.1	51.2	59.9	67.0	74.6	75.5	67.7	59.1	50.6	38.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	56.1	56.2	62.4	70.9	74.7	80.0	79.8	76.2	69.3	62.4	51.7	(27)
(28)			MCD B	48.9	58.1	63.0	72.3	74.7	80.5	86.9	89.0	82.8	72.6	67.3	55.8	(28)
(29)		2%	WB	39.3	47.6	53.5	58.7	66.5	72.8	78.6	78.6	74.2	66.4	58.2	44.9	(29)
(30)			MCD B	44.2	52.1	59.5	69.4	75.7	79.2	86.2	87.5	79.0	70.3	61.3	48.5	(30)
(31)		5%	WB	37.0	42.3	50.5	56.6	64.1	71.0	77.5	77.7	72.6	63.6	54.9	41.8	(31)
(32)			MCD B	41.6	48.1	58.2	65.0	74.0	77.3	84.6	85.8	76.9	66.9	58.7	45.8	(32)
(33)		10%	WB	34.3	39.1	47.1	54.3	62.1	69.4	76.4	76.8	70.6	60.4	52.0	39.1	(33)
(34)			MCD B	39.5	46.1	55.3	63.1	71.7	76.2	83.0	84.1	75.1	65.7	55.8	44.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.0	20.4	22.6	23.6	24.2	18.2	14.2	14.8	17.5	22.1	20.5	18.7	(35)
(36)			MCDBR	23.9	26.6	31.0	32.8	31.3	24.0	18.7	19.9	20.3	24.8	25.1	23.5	(36)
(37)			MCWBR	15.5	16.6	16.1	14.4	11.6	7.9	6.1	6.5	8.3	12.1	14.7	15.8	(37)
(38)		5% WB	MCD BR	20.8	22.3	24.8	23.4	22.2	15.1	15.8	16.4	14.3	15.3	18.1	20.5	(38)
(39)			MCWBR	14.6	16.1	15.8	13.0	11.0	6.4	6.1	6.0	6.8	9.2	12.1	15.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.434	0.485	0.573	0.603	0.614	0.676	0.625	0.579	0.504	0.476	0.453	0.429	(40)	
(41)		taud	1.936	1.858	1.693	1.677	1.691	1.630	1.799	1.899	2.042	1.993	1.973	1.967	(41)	
(42)		Ebn at Noon	227	231	222	225	225	211	221	228	239	232	220	219	(42)	
(43)		Edh at Noon	46	56	72	76	76	81	68	61	50	48	44	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	843	1070	1386	1583	1758	1565	1366	1392	1255	1153	853	752	(44)	
(45)		RadStd	57	91	129	124	169	138	191	188	138	103	99	48	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+122	N/A

Nomenclature: See separate page